

User & Administrator Guide: Dynamic Command Configuration Engine (`axi\_command\_config`)

Document Version: `1.0.0`
Target Audience: `Axpert Developers, System Administrators, Technical QA Engineers`
Applies To: `Axi Command Palette (`Axi_Beta`) v0.4.0+ / Axpert Web 11.4+

1. Overview & Purpose

The **Dynamic Command Configuration Engine** is a metadata-driven navigation architecture in the Axi Command Palette. Historically, adding or modifying commands was done manually. With the introduction of the ``axi_command_config`` table and REST API, **100%** of command navigation is managed directly in the database. System administrators can now manage the command palette through a REST API or directly in the database.

```
graph TD
    User[User Types Command] --> Autocomplete[Command Palette Autocomplete]
    Autocomplete --> Matcher[Config Matcher: command + prompt_options]
    Matcher --> DBConfig[(axi_command_config)]
    DBConfig --> Resolver[URL & Placeholder Resolver: :username, :userroles, :appname]
    Resolver --> Dispatcher[prompt_option_type]
    Dispatcher --> Tstruct[TstructNav[Redirect to Tstruct Form]]
    Dispatcher --> Iview[IviewNav[Redirect to Iview Report]]
    Dispatcher --> DualNav[Tstruct / Iview]
    Dispatcher --> UrlNav[Custom ASPX Page]
    Dispatcher --> FlowNav[Process Builder Flow]
    Dispatcher --> IviewLoad[Iview-to-Iview Load]
```

2. Database Schema Reference (`axi\_command\_config`)

The table ``axi_command_config`` is stored in the application database (PostgreSQL or Oracle).

Column Specifications

Column Name	Data Type	Nullable	Description & Purpose
<code>`config_id`</code>	<code>`VARCHAR(50)`</code>	No	Primary Key. Unique identifier for the configuration rule (e.g. <code>`cfg_configure_user`</code>).
<code>`command`</code>	<code>`VARCHAR(50)`</code>	No	The root command verb (e.g. <code>`Configure`</code> , <code>`SDK`</code> , <code>`Upload`</code> , <code>`Download`</code> , <code>`View`</code> , <code>`Create`</code> , <code>`Edit`</code>).
<code>`prompt_options`</code>	<code>`VARCHAR(200)`</code>	No	The specific prompt option value to match (e.g. <code>`user`</code> , <code>`role listing`</code> , <code>`app`</code> , <code>`default`</code>). Case-insensitive.
<code>`prompt_id`</code>	<code>`VARCHAR(50)`</code>	No	Target identifier: <code>`TransID`</code> for Tstructs, <code>`IviewName`</code> for Iviews, or filename for custom pages.
<code>`prompt_option_type`</code>	<code>`VARCHAR(20)`</code>	No	Execution mode determining how the client shell handles the navigation.
<code>`param_field`</code>	<code>`VARCHAR(100)`</code>	Yes	Query parameter key to bind the user's input parameter to (e.g. <code>`name`</code> , <code>`type`</code> , <code>`actorname`</code>).
<code>`target_url`</code>	<code>`VARCHAR(500)`</code>	Yes	Explicit target URL template override. If omitted, constructed dynamically based on <code>`prompt_option_type`</code> .
<code>`extra_params`</code>	<code>`VARCHAR(500)`</code>	Yes	Additional query string parameters or dynamic tokens to append (e.g. <code>`&status=true&user=:username`</code>).
<code>`active`</code>	<code>`VARCHAR(1)`</code>	Yes	Status flag (<code>`T`</code> for Active, <code>`F`</code> for Disabled). Default: <code>`T`</code> .

3. Supported Navigation Modes (`prompt\_option\_type`)

The engine provides specialized execution handlers tailored to Axpert structure types:

1. `tstruct` (Transaction Forms)

- Use Case:** Opens an Axpert Tstruct form for creating a new record or editing an existing record.
- Behavior:** If user provides a search value and ``param_field`` is specified, it opens in edit mode (``hltype=load``); otherwise opens in create mode (``hltype=create``).
- Example:**

```
sql
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, active)
VALUES ('cfg_configure_user', 'Configure', 'user', 'axusr', 'tstruct', 'uname', 'T');
```

2. `iview` (Interactive Views / Reports)

- Use Case:** Opens an Axpert interactive view report.
- Behavior:** Routes directly to ``iview.aspx?iviewname=``.
- Example:**

```
sql
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, active)
VALUES ('cfg_configure_users', 'Configure', 'user listing', 'axusers', 'iview', 'T');
```

3. `tstruct/iview` or `iview/tstruct` (Dual-Mode Conditional Routing)

- Use Case:** Single command that opens a listing report by default, but directly opens a specific record form if an entity name/ID is supplied.
- Syntax for ``prompt_id``:** ``<entity>/<id>`` (e.g. ``ad_ur/ad___url``).
- Behavior:**
 - If user types ``Configure Role`` $\rightarrow$ Opens Iview report ``ad___url``.
 - If user types ``Configure Role "Admin"`` $\rightarrow$ Opens Tstruct form ``ad_ur`` with record ``Admin``.
- Example:**

```
sql
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, active)
VALUES ('cfg_role_dual', 'Configure', 'role', 'ad_ur/ad___url', 'tstruct/iview', 'rname', 'T');
```

4. `url` (Standard Custom ASPX / HTML Pages)

- Use Case:** Redirects to custom Axpert web pages, tools, or configuration dashboards.
- Behavior:** Resolves relative URLs to ``../aspx/`` or uses explicit ``target_url``.
- Example:**

```
sql
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, target_url, active)
VALUES ('cfg_config_app', 'Configure', 'app', 'appconfig.aspx', 'url', NULL, '../aspx/appconfig.aspx', 'T');
```

5. `processflow` or `url/tstruct` (Process Builder Flow)

- Use Case:** Opens visual workflow builder diagrams.
- Behavior:** Routes to ``processflow.aspx?loadcaption=AxProcessBuilder&processname=``. If no parameter is passed, falls back to ``ad_pm`` Tstruct.

```

- **Example:**
  ```sql
 INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, target_url, active)
 VALUES ('cfg_configure_peg', 'Configure', 'peg', 'ad_pm', 'processflow', 'processname', '../asp/processflow.aspx?loadcaption=AxProcessBuilder', 'T');
  ```

### 6. `ivtoivload` (IView-to-IView Parameterized Navigation)
- **Use Case:** Navigates to parameterized summary IViews with preset filter bindings.
- **Example:**
  ```sql
 INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, param_field, target_url, extra_params, active)
 VALUES ('cfg_configure_dim_list', 'Configure', 'dimension listing', 'ad__upg', 'ivtoivload', 'prole', '../asp/ivtoivload.aspx?ivname=ad__upg', 'AxOpen
  ```

```

4. Dynamic URL Placeholders

You can include contextual placeholders in `target\_url` and `extra\_params`. The client engine resolves these dynamically from the logged-in session before

| Placeholder | Token | Resolved Runtime Value | Description |
|-------------------|----------------------------------|--|-------------|
| ** :username ** | `window.mainUserName` | The login username of the current user. | |
| ** :userroles ** | `window.AxUserRoles` | Comma-separated list of active user roles. | |
| ** :userresp ** | `window.userResp` | Active responsibility of the logged-in user. | |
| ** :appname ** | `appname` / `window.mainProject` | Current application project schema name. | |
| ** :param ** | `paramValue` | The parameter value typed or selected by the user. | |
| ** :paramField ** | `paramField` | The field name configured for the target. | |

Placeholder Usage Example

```

```sql
target_url: "../asp/customdashboard.aspx?user=:username&role=:userroles&proj=:appname"
```

```

5. Step-by-Step Tutorial: Configuring a New Command

Scenario

You want to add a new command: `**Configure SLA "Gold Policy"**` that opens custom page ``SlaConfig.aspx`` with the selected policy name and current user's r

Step 1: Register Autocomplete Prompt

If ``Sla`` is a new sub-option under ``Configure``, ensure it is listed in ``axi_command_prompts`` so users receive auto-complete suggestions when typing ``Config`

```

```sql
-- 1. Append option name to Position 2 (Object Type values)
UPDATE axi_command_prompts
SET promptvalues = promptvalues || ',SLA'
WHERE cmdtoken = 4 AND wordpos = 2;

-- 2. Append matching data source to Position 3 (Object Name data source)
UPDATE axi_command_prompts
SET promptsource = promptsource || ',Axi_Dummy'
WHERE cmdtoken = 4 AND wordpos = 3;
```

```

Step 2: Insert the Configuration Row

Execute the following SQL statement in your database:

```

```sql
INSERT INTO axi_command_config (
 config_id,
 command,
 prompt_options,
 prompt_id,
 prompt_option_type,
 param_field,
 target_url,
 extra_params,
 active
) VALUES (
 'cfg_configure_sla',
 'Configure',
 'sla',
 'SlaConfig.aspx',
 'url',
 'policyname',
 '../asp/SlaConfig.aspx',
 'role=:userroles&user=:username',
 'T'
);
```

```

Step 3: Test in the Command Palette

1. Open Axi Web Shell and press ``Ctrl + Space`` to activate Axi Command Palette.
2. Type: ``Configure SLA "Gold Policy"``
3. Press ``Enter`` or click `**Go**`.
4. The system executes:


```

      ```
 ../asp/SlaConfig.aspx?policyname=Gold%20Policy&role=Developer&user=admin
      ```
    
```
5. Click `**Pop**` to test tabbed execution in ``PopupContainer.html``.

6. Troubleshooting & FAQs

Q1: I added a row in `axi\_command\_config`, but the command palette says "Unknown command".

- ****Check Cache:**** The configuration is loaded when the shell initializes. Refresh the browser (`Ctrl + F5`) or click the ****Refresh**** button on the Axi toolbar.
- ****Check Matching:**** Verify that `command` matches the root verb exactly (e.g. `Configure`) and `prompt\_options` matches the typed sub-verb in lowercase.

Q2: How do I create a fallback route for an entire command group?

- Set `prompt\_options` to `default`. If no exact sub-option matches, the engine falls back to this rule:

```
```sql
INSERT INTO axi_command_config (config_id, command, prompt_options, prompt_id, prompt_option_type, active)
VALUES ('cfg_default_sdk', 'SDK', 'default', 'devexplorer', 'url', 'T');
```

### Q3: My custom page opens in the main window, but I want it to open in a popup tab.

- Users can press `Ctrl + Shift + Enter` or select the **\*\*Pop\*\*** action button. The engine automatically appends `&AxisPop=true` and loads the page within `F

### Q4: Can I temporarily disable a command without deleting the row?

- Yes. Update the status flag: `UPDATE axi\_command\_config SET active = 'F' WHERE config\_id = 'cfg\_configure\_sla';`.